Replication: Subjective Life Expectancies, Time Preference Heterogeneity, and Wealth Inequality
Foltyn & Olsson (2024)

Luca Seibert - 8511270

March 27, 2026

Abstract

This paper replicates the overlapping generations model with idiosyncratic shocks by Foltyn and Olsson (2024), focusing on how subjective survival beliefs generate wealth inequality. I implement the core dynamics of health-dependent survival expectations while simplifying the framework: no stochastic intergenerational bequests, and a partial equilibrium tax system. By calibrating β to match target capital-output ratios, the replication recovers the paper's median wealth dynamics and health-wealth gap despite these omissions.

Contents

1 Introduction

This paper replicates the model in *“Subjective life expectancies, time preference heterogeneity, and wealth inequality”* by Foltyn and Olsson (2024). Their core contribution is a health-dependent bias in survival beliefs: healthy individuals overestimate survival, while those in poor health underestimate it. This bias alters intertemporal consumption-savings decisions and distorts optimal behavior in rational expectation models. The authors estimate an econometric model using HRS data for non-black males, extracting income processes, medical shock distributions, and subjective survival probabilities. These estimates feed into an OLG model with idiosyncratic shocks featuring a 90-period life cycle (45 working, 45 retirement). Thereby, they illustrate that this modeled survival bias explains roughly one-fifth of the observed health-wealth gap.
My Python replication simplifies the original framework. I drop the stochastic intergenerational bequest system and adopt a semi-general equilibrium approach with a partial equilibrium tax system. By adjusting the discount factor, I match the target capital-output ratio and recover the key behavioral mechanism without modeling bequest redistribution.
Section 2 defines the formal model. Section 3 discusses computational implementation. Section 4 presents the replication results.

2 Model Framework

The economic model is an overlapping generations (OLG) framework populated by cohorts of heterogeneous agents facing idiosyncratic shocks. Agents live for a maximum of $T = 90$ model periods, of which they spend the first 45 periods working and starting with model period $T_r = 46$ the remaining 45 periods in retirement. Each new cohort is initially populated by an alive mass of 1.0 and is subject to age-dependent mortality risks.
Throughout their life cycle, the agents' optimization problem is subject to idiosyncratic health, medical, and productivity shocks, as well as transitory shocks. Importantly, the onset of these components is calibrated to match the empirical life-cycle dynamics estimated in the econometric model of the paper. Furthermore, agents receive no income in the death state but remain subject to final out-of-pocket medical expenses. The overall framework is strictly calibrated to the US economy, the details of which will be further illustrated in Section 2.4.

2.1 The Agent's Problem

In every period, agents decide on consumption (c) and savings (s) to maximize their expected lifetime utility. They can save in a riskless bond yielding a constant gross return R, and are constrained by a strict no-borrowing rule ($s \geq 0$).
Because the model does not feature an endogenous labor supply decision, the fundamental optimization problem remains structurally consistent across the working and retirement phases. The primary difference by transitioning into retirement is the income component, that changes from net working income i_{work} to net pension income i_{retire} . Sticking to this notation we can derive a common recursive description of the optimization problem.
Recursive Formulation. The optimization problem at each time period is defined over the state space $\mathbf{z} := (t, x, h, p, \eta)$, where t is age, x is cash-at-hand, h is the health state, p is the persistent productivity state, and η is the persistent medical state.

Let $j \in \{\text{work}, \text{retire}\}$ denote the life-cycle phase, where $j = \text{work}$ for $t \in \{1, 2, .., T_r - 1\}$ and $j = \text{retire}$ for $t \in$: $\{T_r, T_r + 1, .., T\}$. The evolution of cash-at-hand is given by:

$$x' = Rs + i'_j - m(t + 1, h', \eta', \nu') + \zeta' \tag{1}$$

where i'_j refers to either net working income or net pension income in the subsequent period. The term $m(\cdot)$ represents out-of-pocket medical expenses, which depend on health h', the persistent medical component η', and a transitory medical component ν'. The term ζ' represents a government-provided social safety net that ensures a minimum consumption floor $\underline{c}$:

$$\zeta' = \max(0, \underline{c} + m(t+1, h', \eta', \nu') - Rs - i'_j) \tag{2}$$

Whenever an agent requires government transfers ($\zeta' > 0$), they are constrained to consume exactly the floor ($c = \underline{c}$) and save nothing ($s = 0$).
Agents also possess a warm-glow bequest motive, capturing the utility derived from leaving accidental wealth, given by:

$$V_b(b) = \phi_1 \frac{(b + \phi_2)^{1-\sigma} - 1}{1 - \sigma} \tag{3}$$

where ϕ_1 governs the marginal utility of leaving a bequest, and ϕ_2 controls the extent to which bequests act as a luxury good. Here, b' denotes the net bequest left after accounting for final medical expenses in the death state and a government inheritance tax $T_b(\cdot)$:

$$b' = \max\{0, Rs - m(t+1, \eta', \nu')\} - T_b(\max\{0, Rs - m(t+1, \eta', \nu')\}) \tag{4}$$

Therefore, the agent's optimality decision is governed by the following Bellman equation:

$$\begin{aligned} V(\mathbf{z}) = \max_{c,s} & \left\{ \frac{c^{1-\sigma} - 1}{1 - \sigma} + \beta \pi^s_{t,h} \mathbb{E}[V(\mathbf{z}')|t, h, p, \eta] + \beta(1 - \pi^s_{t,h}) \mathbb{E}[V_b(b')|t, h, \eta] \right\} \\ \text{s.t.} \quad & s + c \leq x \\ & s \geq 0 \end{aligned} \tag{5}$$

where $\pi^s_{t,h}$ is the age- and health-dependent survival probability. With the general recursive structure established, we can now rigorously define the regime-dependent income process i_j for workers and retirees.
Workers ($t < T_r$) supply a normalized unit of labor (1.0). Given a deterministic, age-health dependent earnings profile $\omega_{t,h}$, gross labor productivity y during the working phase is:

$$y = \omega_{t,h} p \epsilon \tag{6}$$

where p is the persistent productivity component and ϵ is an additional transitory productivity shock. Labor income is defined by scaling the equilibrium wage w by this productivity realization, subject to both payroll taxes $T_{ss}(\cdot)$ and a progressive income tax scheme $T_y(\cdot)$. Net working income i_{work} is defined as:

$$i_{work} = [y - T_{ss}(y)]w - T_y([y - T_{ss}(y)]w) \tag{7}$$

Retirees ($t \geq T_r$) no longer face labor productivity shocks. Instead, the persistent productivity state p becomes permanently fixed at its final pre-retirement realization, serving as a proxy for lifetime earnings. Retirees earn a gross pension income wy_r, where:

$$y_r = \hat{\omega}_r R_{ss}(p) \tag{8}$$

This pension is a function of the average economy-wide earnings profile in the period prior to retirement, $\hat{\omega}_r$, and a replacement function $R_{ss}(p)$ that mimics the regressive bend-points of the US Social Security system.
Because pension income is only subject to the non-linear income tax $T_y(\cdot)$ and not payroll taxes, net pension income i_{retire} reduces to:

$$i_{retire} = wy_r - T_y(wy_r) \tag{9}$$

2.2 Technology

The production side features a standard Cobb-Douglas aggregate production function with competitive firms, given by $F(A, K, L) = AK^{\alpha}L^{1-\alpha}$. Capital depreciates at a constant rate δ. I follow the approach of the authors by adjusting the total factor productivity (A) in equilibrium such that the equilibrium wage w is fixed to unity.

2.3 Government and General Equilibrium

I pursue a semi-general equilibrium approach to focus strictly on the agents' belief biases. Because the scope of this replication does not involve analyzing reforms to the social security system, this simplification isolates the core mechanism of subjective versus objective health beliefs. Note, that I still mirror the tax system implemented in the paper by calibrating the tax parameters at their General Equilibrium reports. The government runs a balanced fiscal scheme behind the scenes. It collects income taxes from working agents, taxes accidental bequests, and finances the consumption floor. These taxes are defined as:

$$T_{ss}(y) = \tau_{ss} \min\{y, y_{max}\} \tag{10}$$

$$T_y(i) = i - \lambda i^{1-\tau} \tag{11}$$

$$T_b(x) = \begin{cases} 0 & \text{if } x \leq \chi_b \\ \tau_b(x - \chi_b) & \text{otherwise} \end{cases} \tag{12}$$

As intergenerational inheritances are not explicitly redistributed to households in this baseline framework, the government collects accidental bequests to finance the consumption floor and other aggregate spending.
The regressive replacement rate of the US Social Security system, $R_{ss}(p)$, is defined mathematically as:

$$R_{ss}(p) = \begin{cases} \rho_1 p & \text{if } p \leq p_1^* \\ \rho_1 p_1^* + \rho_2(p - p_1^*) & \text{if } p_1^* < p \leq p_2^* \\ \rho_1 p_1^* + \rho_2(p_2^* - p_1^*) + \rho_3(\min\{p_{max}^*, p\} - p_2^*) & \text{otherwise} \end{cases} \tag{13}$$

where the bend points p_1^*, p_2^*, and the maximum taxable earnings cap p_{max}^* are calibrated to the US economy and fully expressed in internal model units:

$$p_i^* = \frac{(b_i^{\$}/e_{med}^{\$})y_{med}}{\hat{\omega}_r}, \quad p_{max}^* = \frac{(e_{max}^{\$}/e_{med}^{\$})y_{med}}{\hat{\omega}_r} \tag{14}$$

where $e^{\$}$ and $b^{\$}$ correspond to the empirical median income and bend points calibrated to the US pension system, and $\hat{\omega}_r$ is the average earnings profile in the period prior to retirement.

2.4 External Calibration and Model Normalization

Preferences

Following the original implementation, the coefficient of relative risk aversion is set to $\sigma = 1$, implying a logarithmic utility function $u(c) = \log(c)$. The discount factor is calibrated to $\beta = 0.986$ to obtain a Capital to Output ratio of approximately 3.0. Note that the paper uses $\beta = 0.979$. For the baseline scenario, the intentional bequest motive is shut down by setting $\phi_1 = 0$, meaning any wealth left at death is purely accidental.

Demographics

Agents enter the economy at age 20 (model period 0) and die with certainty at age 109 (model period $N_T = 89$). Mortality risks commence at model period 30 (age 50). Survival transitions $\pi^s_{t,h}$ are taken directly from the econometric estimates of Foltyn and Olsson (2024). For the scenario without health heterogeneity (NHH), I utilize the average objective transition probabilities adjusted for the ergodic health process, ensuring that aggregate mortality remains identical to the objective health heterogeneity scenario.

Earnings

The age-health dependent productivity profile, $\omega_{t,h}$, is central to calibrating the model to the US economy. Utilizing the profile estimated by the authors, this deterministic process is additionally scaled by persistent and transitory idiosyncratic shocks. The overall labor productivity process is calculated as:

$$\log y_{t,h} = \log \omega_{t,h} + \log p_t + \log \epsilon_t \quad \text{for } t \in \{1, \ldots, T_r - 1\} \tag{15}$$

where the persistent component (p_t) follows an AR(1) process:

$$\log p_t = \rho_p \log p_{t-1} + v_t \tag{16}$$

with autocorrelation ρ_p and i.i.d. innovations $v_t \sim \mathcal{N}(0, \sigma^2_v)$. The transitory shock ($\epsilon_t$) is i.i.d. log-normally distributed such that $\log \epsilon_t \sim \mathcal{N}(0, \sigma^2_\epsilon)$. I adopt the authors' calibration, setting $(\rho_p, \sigma^2_v, \sigma^2_\epsilon) = (0.9695, 0.0384, 0.0522)$. To match the moments of the highly persistent productivity shock, I discretize the AR(1) process using the Rouwenhorst method. The log-normal transitory process is discretized using the Tauchen method with $\rho = 0$.

Medical Expenditures

Central to the underlying study are the age- and health-dependent medical expenses. I adopt the authors' econometric estimates to model the expenditure shocks as follows:

$$\log m_{t,h} = \mu_{t,h} + \sigma_{t,h}(\eta_t + \nu_t) \tag{17}$$

where $\mu_{t,h}$ and $\sigma_{t,h}$ are the calibrated moments of the expenditure process conditional on age and health. Note that medical shocks are only present for model ages $t \in \{30, 31, 32, ..., T\}$ and thus the state space is fundamentally asymmetric. The persistent medical component (η_t) follows an AR(1) process:

$$\eta_t = \rho_m \eta_{t-1} + \gamma_t \tag{18}$$

where $\gamma_t \sim \mathcal{N}(0, \sigma^2_\gamma)$ and the transitory component is $\nu_t \sim \mathcal{N}(0, \sigma^2_\nu)$. The authors utilize a Generalized Method of Moments (GMM) and Simulated Method of Moments (SMM) procedure to estimate the implied parameter values at an annual frequency. Following their baseline, I set $(\rho_m, \sigma^2_\gamma, \sigma^2_\nu) = (0.920, 0.084, 0.457)$. Similar to the earnings process, I apply the Rouwenhorst method to discretize the persistent medical component and the Tauchen method ($\rho = 0$) for the transitory component.

Bequests

As already mentioned, I omit the stochastic intergenerational bequest system, meaning that agents do not receive inheritances over their lifetime. However, because agents may still derive warm-glow utility from leaving wealth, the bequest motive $V_b(b)$ is maintained in the theoretical model. Even though the government absorbs all accidental bequests without redistributing them to households, the introduction of the estate tax remains necessary, as it distorts the optimal savings decisions of agents' bequest decisions.

Model Normalization

To ensure the model accurately reflects the US economy, I utilize a one-time auxiliary forward simulation of the exogenous processes. This simulation yields the ergodic distribution of labor productivity during the working age, providing the model's median (y_{med}) and average (y_{avg}) income.
I normalize all income-related model components by dividing them by the ergodic gross labor productivity average, y_{avg}. This scales the internal unit of account such that average ergodic income equals unity (1.0). The authors report a US dollar consumption floor of \$2,325, which is explicitly calibrated to represent 5% of annualized average earnings. In my normalized model, this corresponds to a consumption floor of $\underline{c} = 0.05$. Therefore, the implied scaling factor to convert normalized model units back into dollar-denominated values is $\$2,325/0.05 = \$46,500$.

Remaining Externally Calibrated Parameters.

Other parameters used in the model are presented in Table 1 and are strictly chosen to represent the US social security system and to represent the government tax system in the general equilibrium structure of the paper.

Table 1: Calibrated Parameter Values

Parameter	Description	Value	Source
Technology			
α	Capital share	0.36	Foltyn and Olsson (2024)
δ	Depreciation rate	0.096	Foltyn and Olsson (2024)
A	TFP	0.891	Targeting $w = 1.0$ (Orig: 0.896)
β	Time preference	0.986	Targeting $K/Y \approx 3.0$
Social Security			
ρ_1, ρ_2, ρ_3	Replacement rates	0.90, 0.32, 0.15	Foltyn and Olsson (2024)
b_1, b_2	Bend points (USD)	\$6,384, \$38,424	Foltyn and Olsson (2024)
y_{avg}	Pre-normalization avg. productivity	39.564	Ergodic Simulation
$\underline{c}$	Consumption floor	\$2,325	5% of Avg. Earnings
Fiscal Policy			
τ	Tax progressivity	0.137	Foltyn and Olsson (2024)
λ	Tax level	0.9222	Foltyn and Olsson (2024)
τ_{ss}	Payroll tax rate	0.124	Social Security Administration (2024)

2.5 Model Definitions

The core contribution of the paper is to introduce a calibrated bias in survival expectations that distorts agents' consumption-savings decisions. Therefore, the paper defines three setups of the model discussed in Section 2.1, utilizing their calibrated objective "true" survival probabilities $\pi_{t,h}$ and estimated subjective survival probabilities $\tilde{\pi}_{t,h}$.

Subjective Survival Heterogeneity (SSH).

This is the benchmark economy. Agents experience idiosyncratic health shocks and solve their optimal consumption-savings policy using the subjective survival probabilities ($\tilde{\pi}_{t,h}$). However, the actual forward simulation of the population mass iterates according to the "true" objective survival probabilities ($\pi_{t,h}$). This discrepancy creates the *survival bias*, capturing the empirical under- and overestimation of mortality risks reported in the paper.

Objective Survival Heterogeneity (OSH).

This is the rational expectations benchmark. Agents are fully aware of their true, objective survival probabilities, meaning their internal optimal choices align perfectly with the transition probabilities of the forward simulation.

No Health Heterogeneity (NHH).

Health heterogeneity is removed entirely in this scenario. All health-dependent components are collapsed into an ergodic average such that the NHH survival probabilities equal the cross-sectional mean of the objective probabilities ($\hat{\pi}_t = \sum_h \pi_{t,h}\mu(h)$). Consistent with the Python implementation, this marginalization over the ergodic distribution is also applied to medical expenditures and labor earnings to fully collapse the health dimension.

Cross-Economy Normalization.

To compare economies in the same units, the SSH equilibrium serves as the anchor. I first solve SSH general equilibrium to pin down Total Factor Productivity (A) such that $w = 1$ and extract the model scaling factor y_{avg}. OSH and NHH then use this anchored A, allowing their wages to adjust endogenously.

3 Computational Challenges and Solution Strategy

Solving the agent's life-cycle problem in this framework is highly non-trivial. The combination of a high-dimensional state space, the non-concavity induced by the consumption floor, and the requirement for structural flexibility to accommodate asymmetric life-cycle shocks introduces three primary computational bottlenecks.

3.1 Computational Challenges

The Curse of Dimensionality

As the model embeds persistent medical and productivity shocks within a 90-period OLG framework, the state space grows exceptionally large. As illustrated in Table 2, the individual state space consists of five dimensions with a total of approximately 2.7 million grid points. Notably, the total number of grid points is not the product of the individual grid sizes, because of the asymmetric shocks over the life cycle. Evaluating the value function over this space while integrating over transitory shock states demands a highly efficient numerical implementation.

Dimension	**Size**
Individual State Space	
Cash-at-hand (x)	200
Age (t)	90
Health (h)	6
Persistent labor productivity (p)	5
Persistent medical expenditures (η)	7
Total number of grid points	$\approx$ 2.7M
Transitory Shocks	
Labor productivity (ϵ)	3
Medical expenditures (ν)	5

Table 2: Grid Dimensions for the Household Problem

Because this replication is carried out in Python, overcoming the lack of native compilation speed was paramount. To achieve high-performance execution, Python classes handle high-level data management and memory allocation, while all computationally intensive loops are outsourced to Just-In-Time (JIT) compiled functions using the `Numba` library (Lam, Pitrou, and Seibert 2015). This allows the model to benefit from Python's expressive data structures while achieving near-compiled execution speed in the innermost loops.

State-Space Asymmetry and Counterfactual Flexibility

A core computational requirement is the ability to toggle between subjective, objective, and no-health-heterogeneity scenarios. However, the structure of the optimization problem is not uniform across the life cycle: medical shocks enter only at age 50, productivity shocks cease at retirement, and income sources shift from labor earnings to pensions. Thereby three distinct model periods can be classified as:

1. **Working life without medical shocks** (Ages 1–30)
2. **Working life with medical shocks** (Ages 31–45)

3. **Retirement phase** (Ages 46–110)

Naively, this would require separate solver routines for each life phase. To avoid this, age can be treated as a *hyper-space variable* that pins down the exact problem structure at each period of life. At each step of the backward induction, an "Age-Problem Constructor" reads the current period and returns the appropriate transition matrices, income components, and medical expenditure grids. To implement a robust routine, I fix the shape of value and policy functions across all periods. This fixed structure comes at a cost: during the first 30 years, when medical shocks are absent, the solver still iterates over the medical dimension.

We can still exploit the asymmetric structure by noting that transitory shocks do not affect the shape of the value function, yet integration over them is unnecessary when the corresponding process is inactive. By using age again as the problem definition, I set the discretized weights of inactive transitory shocks to 1.0, so that integration collapses to a single evaluation. The same logic applies to the retirement phase, where productivity no longer affects pension income. Here, the Markov transition matrix for productivity is set to the identity matrix, freezing the state at its last pre-retirement value, consistent with the pension system depending on terminal working-life productivity.

The last issue remaining is then the different shape of the health states when changing between Health Heterogeneity (SSH / OSH) and No Health Heterogeneity (NHH). This is solved with a similar idea, by letting the shape of the transition matrix determine the structure of the solver. Depending on the length of the health Markov matrix, the code determines automatically what scenario is computed and adjusts accordingly by iterating over the present health states.

Non-Concavity of the Value Function.

The interaction of the borrowing constraint, the consumption floor, and the bequest motive, particularly paired with the cliff tax on estates, produces a non-concave value function. Consequently, standard Euler-equation methods or optimization routines that rely on monotonicity or differentiability cannot be reliably applied. Although there are extensions of Euler methods that relax these conditions (see Fella (2014)), this replication follows the robust discrete grid search approach of the original paper.

While discrete grid search is robust enough to handle non-concavities, its brute-force nature is inherently inefficient, which is particularly problematic in the high-dimensional model environment. To mitigate this, Foltyn and Olsson (2024) utilize a dynamic grid approach, where the search points are dynamically reconstructed around the values identified in the previous iteration of the general equilibrium loop, exploiting the contraction mapping structure of the problem.

To navigate these non-concavities, I instead opt for a static, hybrid approach. The baseline discrete search is augmented with a Golden Section Search (GSS) algorithm. Specifically, the solver performs a sparse coarse grid scan to identify the local region likely containing the global maximum. The GSS then refines the solution within this bracket to achieve high precision without the memory overhead of a dense global grid.

A robustness check ensures that the refined GSS solution is only accepted if its value exceeds the highest value found during the initial coarse scan, preventing the algorithm from being trapped in local maxima. Note that immediately utilizing GSS is not possible as the non-concavity of the problem generates multiple local maxima, and a standalone GSS routine would likely converge to a suboptimal local solution. By bracketing the search via the coarse scan, the hybrid method preserves the robustness of discrete search while achieving the precision of continuous optimization.

Note at this stage, that it is paramount to collapse the recursive structure in Equation 5 into a one-dimensional optimization routine. By using the binding borrowing constraint, we can rewrite the Bellman equation as:

$$V(\mathbf{z}) = \max_{s\in[0,x]} \left\{ \frac{(x-s)^{1-\sigma}-1}{1-\sigma} + \beta\pi^{s}_{t,h}\mathbb{E}[V(\mathbf{z}')|t,h,p,\eta] + \beta(1-\pi^{s}_{t,h})\mathbb{E}[V_b(b')|t,h,\eta] \right\} \tag{19}$$

This collapses the maximization step of the Bellman operator into one single choice variable, s. Moreover, by discretizing the expectations we can separate the model into a current payoff and a continuation value. This step, that I will elaborate more on in the next chapter, is paramount to reduce the problem into a 1-dimensional maximization step, which can then be effectively carried out by the hybrid method mentioned.

Data Management and Calibration

A less obvious challenge was data management. Correctly initializing the life cycle to extract calibration targets like average pre-retirement income, and matching the original authors' scaling, required a careful approach. While I initially intended to omit the tax system entirely, without taxes, the model could not properly balance income against medical expenditure shocks.
Two Python scripts handle the data pipeline and enable the flexible architecture discussed above. The first loads the authors' econometric estimates and exposes all belief modes and grids to the solver. The second, `ModelPrimitives`, reads a config file and constructs the full dimensional problem by discretization, scaling, and age-specific environments, so that the numerical routines remain clean.

3.2 Household Solver: Solution Strategy

Let me now spend more words on the actual numerical implementation and algorithms used to solve the problems mentioned strategically.

High Level Idea

The household solver addresses all discussed bottlenecks simultaneously. The central idea is to decompose the Bellman equation into two separable steps, an *Expectation step* (E-step) and a *Maximization step* (M-step), which together collapse the high-dimensional problem into a single one-dimensional maximization. The backward induction calls a helper function that constructs the exact age-specific environment and pipes the result into a highly optimized solver kernel.
A key simplification follows from observing that cash-on-hand is fully determined given savings s and realizations of (h,p,η):

$$x = (1+r)\,s + y_t(h,p,\epsilon) - m_t(h,\eta,\nu)$$

Thus, we need to only iterate over the exogenous states to imply the cash-at-hand state variable. Let $\mathcal{F} := \mathcal{H}\times\mathcal{P}\times\mathcal{M}$ with typical element $\mathbf{f} = (h,p,\eta)$. Since these processes are independent, iterating over $\mathcal{F}$ is *embarrassingly parallel*. The full procedure is summarized in Algorithm 1.

Hybrid GSS Optimization.

The M-step can be improved by combining a coarse grid scan with Golden Section Search (GSS). Rather than evaluating every point on the savings grid, I first scan a strided subset (every k-th point) to locate the approximate maximizer ψ^*. This coarse scan is performed on the original dense grid, as I require a dense grid for the forward simulation. Therefore, striding the grid leaves me using only one savings grid, which allows a cleaner code structure.
The coarse solution is then refined by bracketing the interval $[\mathcal{S}_{\psi^*-k},\, \min(\mathcal{S}_{\psi^*+k}, x)]$ and applying GSS within this bracket. Because the bracket width equals $2k$ grid spacings, the method

Algorithm 1 Household Solver (Backward Induction)

Require: Grids $\mathcal{X}$, $\mathcal{S}$; Exogenous states $\mathcal{F}$; Transitory shocks $\mathcal{E}$, $\mathcal{V}$, r, β

Terminal: $V_T(x),\ s^*_T(x) \leftarrow \max_s\{U(x-s)+V_b(b(s))\}$
for $t = T-1$ **downto** 1 **do**
  Extract $\Pi_t(\mathbf{f}'|\mathbf{f})$, survival $\pi^s_{t,h}$, income i_{t+1}, medical costs m_{t+1}
  for all $\mathbf{f} \in \mathcal{F}$ **parallel do** ▷ Iterate over exogenous states today
    — *E-Step* —
    Initialize $W(s', \mathbf{f}) \leftarrow 0$ for all $s' \in \mathcal{S}$
    for all $\mathbf{f}' \in \mathcal{F}$, $\epsilon \in \mathcal{E}$, $\nu \in \mathcal{V}$ **do** ▷ Iterate over exogenous states tomorrow
      $x' \leftarrow (1+r)s' + i_{t+1}(\mathbf{f}', \epsilon) - m_{t+1}(\mathbf{f}', \nu)$ ▷ Extract implied cash-at-hand
      $\Omega \leftarrow \pi^s_{t,h} \cdot \Pi(\mathbf{f}'|\mathbf{f}) \cdot \Pr(\epsilon) \cdot \Pr(\nu)$ ▷ Compute joint probability
      $W(s', \mathbf{f}) \mathrel{+}= \beta\,\Omega\,V_{t+1}(x', \mathbf{f}')$ ▷ Interpolate V_{t+1} at x'
    end for
    — *M-Step* —
    for all $x \in \mathcal{X}$ **do**
      $V_t(x, \mathbf{f}),\ s^*_t(x, \mathbf{f}) \leftarrow \max_{s'}\big\{u(x-s') + W(s', \mathbf{f})\big\}$ ▷ Maximize over 1d savings
    end for
  end for
end for
return $\{V_t, s^*_t\}_{t=1}^T$

scans over the entire original grid and considers all values that pure discrete search would. If GSS returns a lower value than the coarse scan the algorithm falls back to the coarse solution.

Algorithm 2 Hybrid M-Step (Coarse Scan + GSS)

Require: Cash-at-hand x, Continuation value W, Savings grid $\mathcal{S}$, Stride k

Coarse Scan: Evaluate every k-th point; store best v^*_{coarse}, index ψ^*
Bracket: $a \leftarrow \mathcal{S}[\max(0, \psi^* - k)]$, $\quad b \leftarrow \min(\mathcal{S}[\psi^* + k],\, x)$
Refine: $v^*_{gss}, s^*_{gss} \leftarrow \mathrm{GSS}(x, W, a, b)$
return $\max(v^*_{coarse}, v^*_{gss})$ and corresponding s^*

3.3 General Equilibrium Solution Strategy

Structure

The general equilibrium builds on the household solver and forward simulation as two modular components. Because the model excludes bequest redistribution and uses calibrated tax data, the equilibrium is best described as a *semi-general equilibrium*: government policy and bequest redistribution remain exogenous, and the only endogenously clearing market is the capital market. On the firm side, total factor productivity A is normalized so that the equilibrium wage is fixed at $w = 1$. The equilibrium is then characterized by a single unknown: the interest rate r^* that clears the capital market.

Forward Simulation

Starting from a newborn cohort at model age 1, the histogram method propagates probability mass forward through the life cycle one period at a time. At each age t, the distribution $\Gamma_t(s, \mathbf{f})$ assigns mass to each point in the state space. For each state with positive mass, cash-on-hand is computed by integrating over transitory shocks, and the policy function yields optimal savings s' via interpolation. Because s' generally falls between grid points, I employ a lottery mechanism: for $\mathcal{S}_j \leq s' \leq \mathcal{S}_{j+1}$, mass is split with weight $\omega = (\mathcal{S}_{j+1} - s')/(\mathcal{S}_{j+1} - \mathcal{S}_j)$ to the lower point and $(1 - \omega)$ to the upper. This ensures a consistent mapping back onto the asset grid. Agents transitioning into the death state contribute their savings as accidental bequests, which are harvested but not redistributed. The procedure is summarized in Algorithm 3.

Algorithm 3 Forward Simulation (Histogram Method)

Require: Policy functions $\{s_t^*\}$, Savings grid $\mathcal{S}$, Exogenous states $\mathcal{F}$, Transitory shocks $\mathcal{E}$, $\mathcal{V}$

Initialize $\Gamma_1(s, \mathbf{f})$ with full mass at $s = 0$ and invariant distribution over $\mathbf{f}$
for $t = 1$ **to** T **do**
    Extract $\Pi_t(\mathbf{f}'|\mathbf{f})$, survival $\pi^s_{t,h}$, income i_t, medical costs m_t
    for all $(s, \mathbf{f})$ **where** $\Gamma_t(s, \mathbf{f}) > 0$ **do**
        — *Transitory Integration & Lottery* —
        for all $\epsilon \in \mathcal{E}$, $\nu \in \mathcal{V}$ **do**
            $x \leftarrow s(1 + r) + i_t(\mathbf{f}, \epsilon) - m_t(\mathbf{f}, \nu)$ ▷ Compute implied cash-at-hand
            $s' \leftarrow s_t^*(x, \mathbf{f})$ ▷ Interpolate policy
            Buffer lottery destinations and weights at s' with mass $\Gamma_t \cdot \Pr(\epsilon) \cdot \Pr(\nu)$
        end for
        — *Persistent State Transitions* —
        for all $\mathbf{f}' \in \mathcal{F}$ **do**
            Map mass onto next period grid $\Gamma_{t+1}(\cdot, \mathbf{f}')$ weighted by $\pi^s_{t,h} \cdot \Pi(\mathbf{f}'|\mathbf{f})$
        end for
    end for
end for
return Distribution history $\{\Gamma_t\}_{t=1}^T$

Even though Γ_t is not defined over cash-at-hand, the cross-sectional distribution of cash-at-hand at any age can be recovered by mapping each state $(s, \mathbf{f})$ through the budget identity $x = Rs + i(\mathbf{f}, \epsilon) - m(\mathbf{f}, \nu)$, integrating over transitory shocks, and weighting by the population mass Γ_t. This reconstruction is used in Section 4.3 to compute the cash-at-hand deciles underlying the savings rate decomposition.

Note at this stage that the idea of the auxiliary forward iteration of Section 2.4 builds upon the same idea, only iterating over the health and productivity shocks, which facilitates the implementation as the savings lottery can be dropped.

Market Clearing

The equilibrium requires aggregating capital and labor supply from the simulated cross-sectional distribution Γ_t. Capital supply sums savings across all ages and states, weighted by the ergodic mass at each point. Labor supply sums productivity across working-age agents, again weighted by ergodic mass, yielding aggregate efficiency units:

$$K^s = \sum_{t=0}^{T} \sum_{s,\mathbf{f}} \Gamma_t(s, \mathbf{f}) \cdot s, \qquad L = \sum_{t=0}^{t_{ret}} \sum_{\mathbf{f}} \Gamma_t(\mathbf{f}) \cdot z_{prod}(\mathbf{f})$$

where $z_{prod,t}(\mathbf{f}) = \omega_{t,h} \cdot p$ denotes efficiency units for a worker in state $\mathbf{f}$ at age t.
For the SSH economy, I normalize the wage to unity and let TFP adjust via the firm FOC. In the OSH and NHH counterfactual, I anchor TFP at the SSH solution and let wages adjust. In both cases, the equilibrium interest rate r^* solves $K^s/L = K^d(r^*)$. The procedure is summarized in Algorithm 4.

Algorithm 4 General Equilibrium (Capital Market Clearing)

Require: Interest rate bounds $[r_{min}, r_{max}]$, tolerance τ
function ComputeGap(r)
    — *Firm Side* —
    if Economy is SSH **then**
        Set $w \leftarrow 1.0$, and impute A and K^d from firm FOCs given r
    else
        Fix $A \leftarrow A_{SSH}$, and compute w and K^d from firm FOCs given r, A
    end if
    Recompute net income grids given current w
    — *Household Side* —
    $\{s_t^*\}, \{V_t\} \leftarrow$ HHSolver(r, w) ▷ Algorithm 1
    $\{\Gamma_t\} \leftarrow$ ForwardSim($\{s_t^*\}$) ▷ Algorithm 3
    — *Aggregation* —
    $K^s \leftarrow \sum_t \sum_{s,\mathbf{f}} \Gamma_t(s, \mathbf{f}) \cdot s$ ▷ Aggregate savings weighted by cross-sectional mass
    $L \leftarrow \sum_{t \leq t_{ret}} \sum_{\mathbf{f}} \Gamma_t(\mathbf{f}) \cdot z_{prod}(\mathbf{f})$ ▷ Aggregate efficiency units over working-age mass
    return $K^s/L - K^d$
end function
$r^* \leftarrow$ Brent(ComputeGap, r_{min}, r_{max}, τ)
return r^*, w^*, A

---

4 Results

This section summarizes the computational and economic results of the replication. I first validate the exogenous model inputs and then present the core replication of median wealth dynamics and the savings belief bias. This is followed by the computational contribution of the hybrid GSS method and a bequest extension. All life-cycle profiles and aggregates reported below are computed by marginalizing over the ergodic distribution Γ_t, ensuring that statistics reflect the actual cross-sectional weights in the economy.

4.1 Exogenous Input Validation

Income and Medical Expense Process.

As we have spent considerable time discussing the interplay of working income, pension income, and medical expenses, let me briefly visualize the calibration set up for the model. Figure 1 illustrates the net model components used in the numerical solution implementation. Note that these are actual averages, not weighted by the ergodic distribution. As a comparison, the authors report their econometric estimates that I used for the model computation. In both cases, the hump-shaped dynamics by health state are clearly visible. Medical expenses appear to be rather small on average; however, note that they are massively skewed, leading to the precautionary savings motive that drives the results.
As a benchmark, Figure 2 illustrates the health dependent econometric estimates the model is based on.

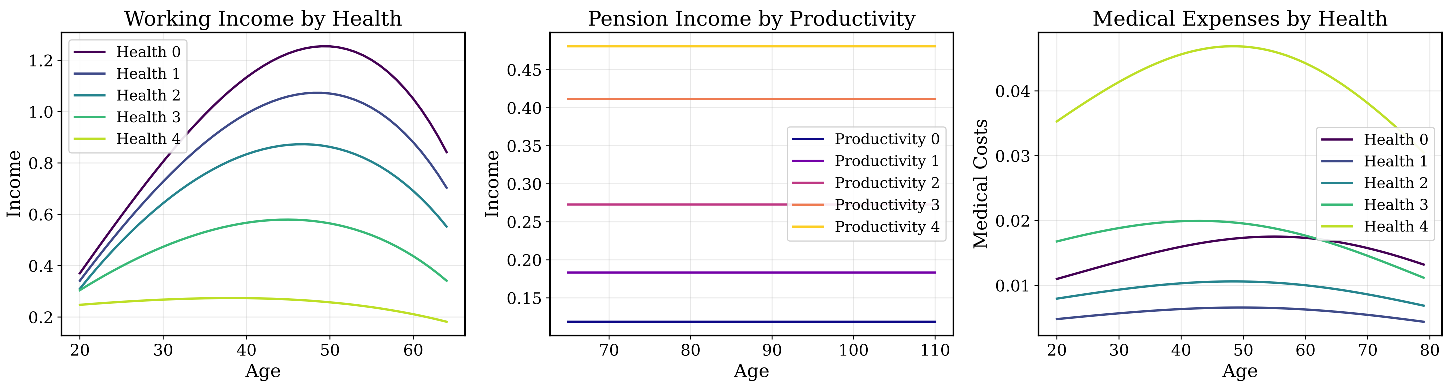


Figure 1: Lifecycle profiles of income and medical expenditures by health state. Left: working income by health. Centre: pension income by productivity quintile. Right: average medical expenses by health. Best Health state indicated by 0, best productivity state by 4

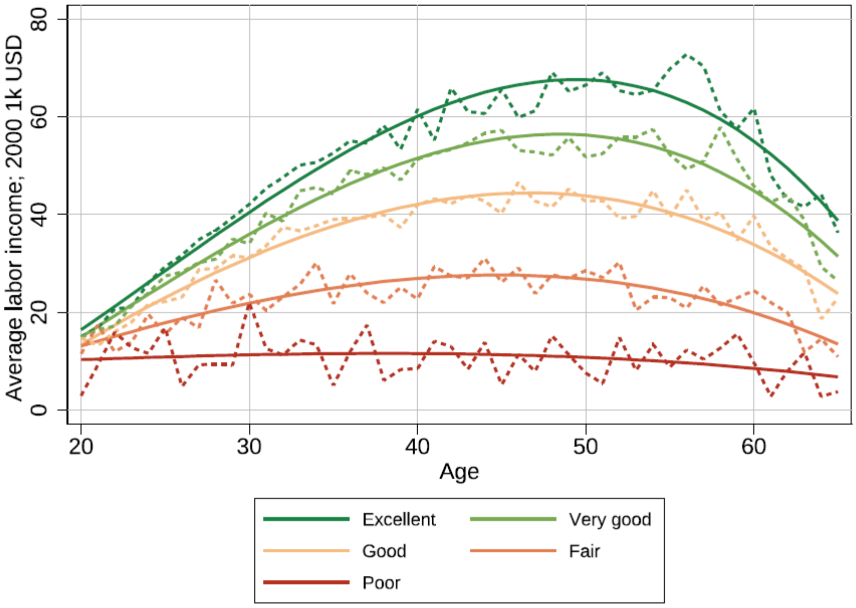
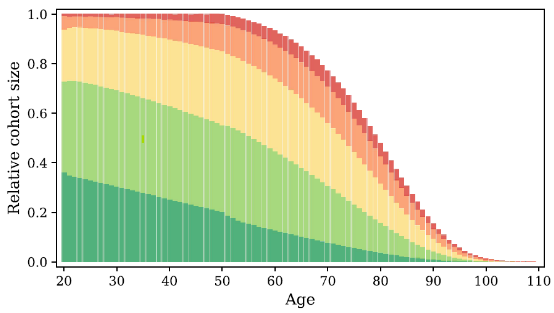


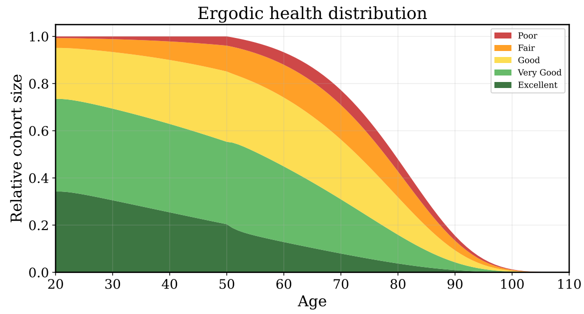
Figure 2: Labor Earnings by health state, nonblack males. Dashed lines indicate raw (weighted) averages by health and age. Solid lines indicate a fitted third-order polynomial. Colors indicate the health state: dark green is excellent while red is poor health. Foltyn and Olsson (2024)

Ergodic Health Distribution

One key component of the model are the estimated survival probabilities. As the calibration by the authors also reports the true econometric survival probabilities used in the forward simulation, a closer look at them is helpful. Figure 3 illustrates the ergodic health states reported in the paper as well as my matching implementation. The key components to highlight are that over the life cycle the relative distribution of health shifts towards poorer states. Also, notice the clear kink in the transition at age 50, at which mortality risks begin and the relative cohort size begins to decreases, reaching approximately zero in the last period. This confirms that the survival rates and the model periods are well calibrated so that the mass of agents surviving to the last model period is negligible.



(a) Original: Foltyn and Olsson (2024)



(b) Replication: Python implementation

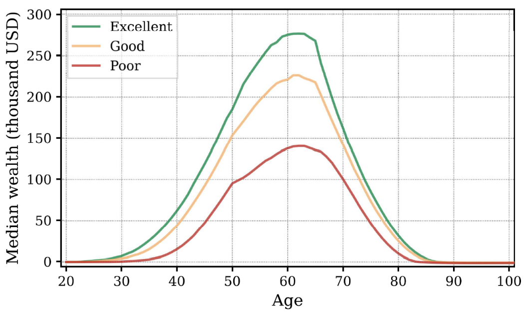
Figure 3: Relative cohort size and health state distribution. Colors indicate the health state: dark green is excellent while red is poor health.

4.2 Median Wealth Dynamics

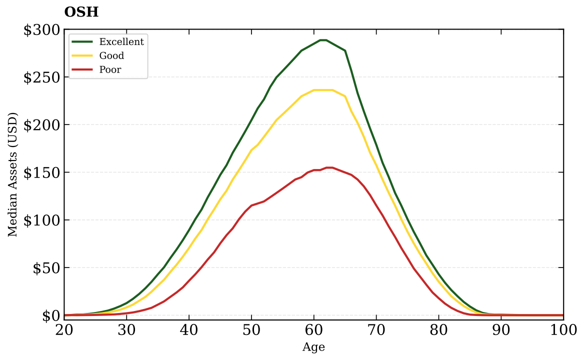
Figure 4 compares median wealth trajectories across the three belief economies: Subjective (SSH), Objective (OSH), and No Health Heterogeneity (NHH). Following the authors, I report results for three health states: Excellent, Good, and Poor.

Rescaling model units by the factor $46,500 (Section 2.4) converts to dollar-calibrated values. The replication is successful: all three economies display the characteristic hump-shaped wealth dynamics, with health-dependent divergence around age 50 when medical shocks and mortality risk activate. The peak timing, relative ordering by health state, and dissaving patterns in retirement all match the original paper.

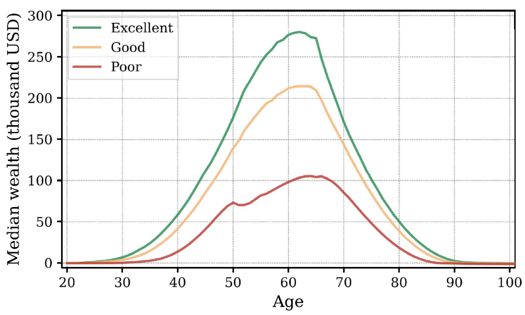
The missing stochastic bequest redistribution, which would inject wealth throughout the life cycle, is compensated by a slightly higher discount factor ($\beta = 0.986$ vs. 0.979 and more discussed in Section 4.4). This raises patience and mimics the savings behavior without the intergenerational transfer mechanism.



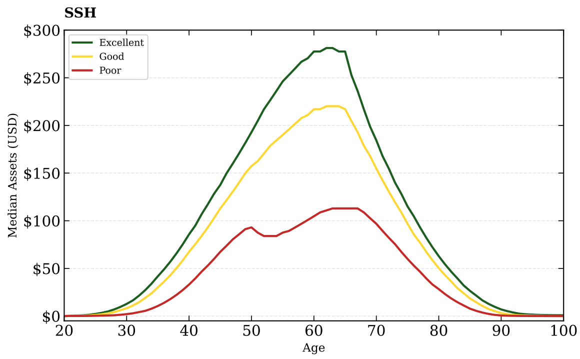
(a) OSH – Paper



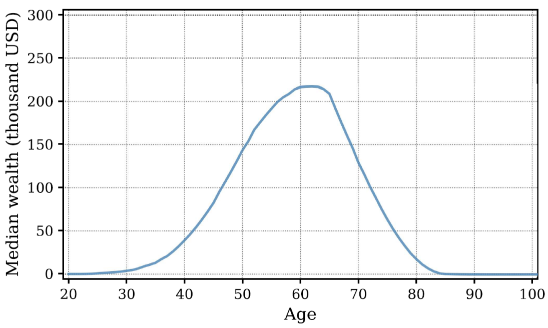
(b) OSH – Replication



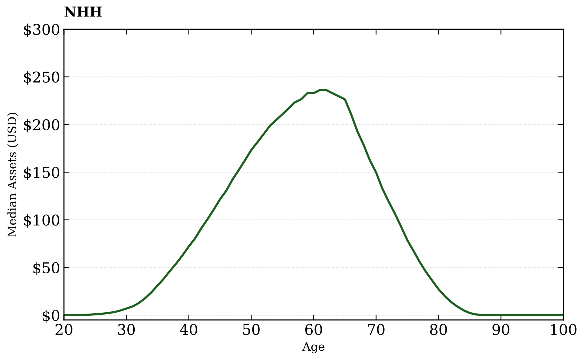
(c) SSH – Paper



(d) SSH – Replication



(e) NHH – Paper



(f) NHH – Replication

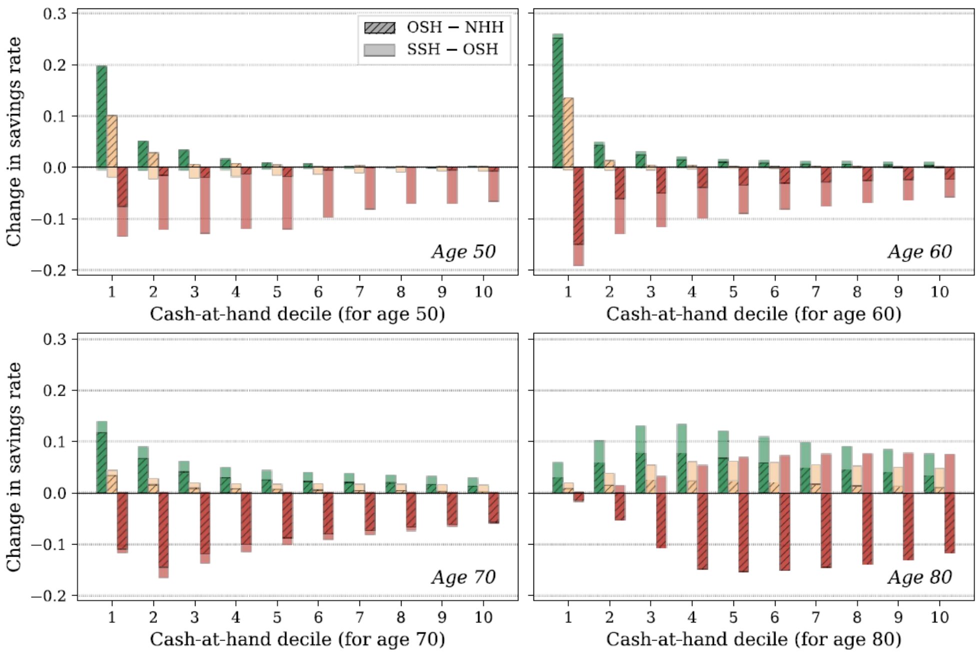
Figure 4: Median Wealth by Age and Health Status. Left column shows original paper; right column shows replication results.

4.3 Savings Belief Bias

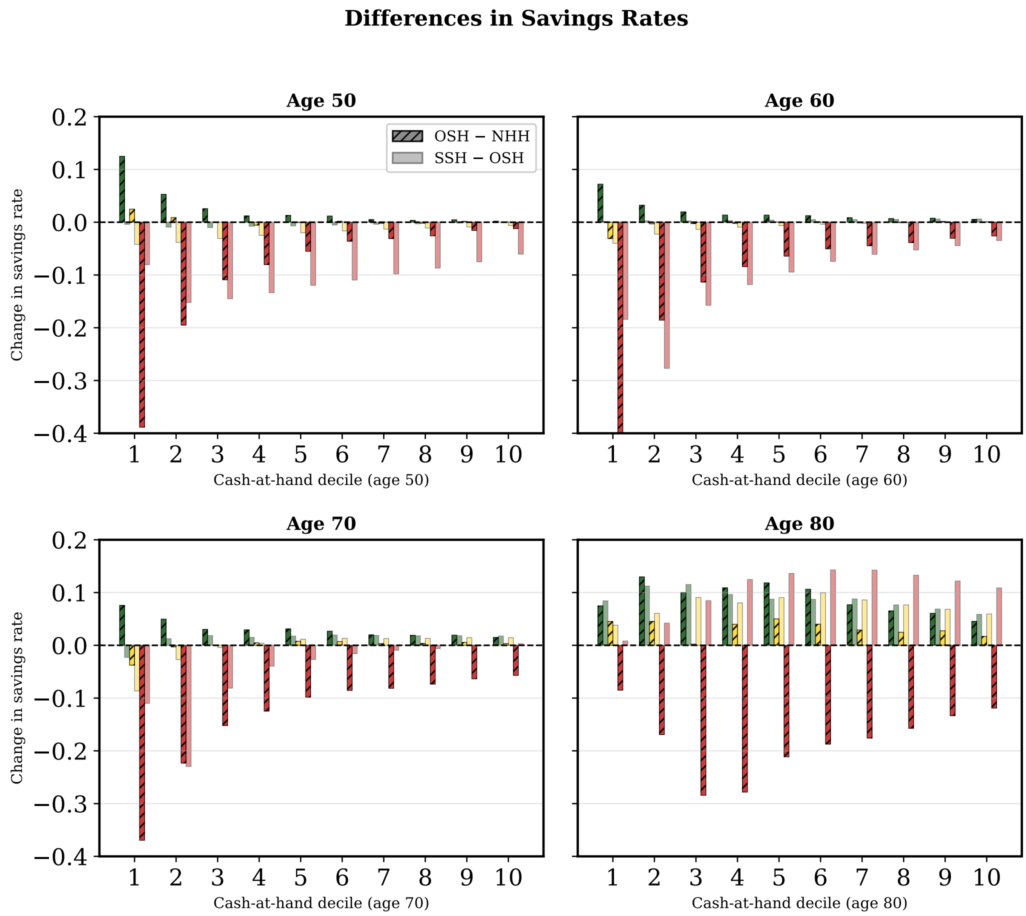
The core result of the paper is the savings bias introduced by the different health groups. Thus, a key figure is the decomposition in Figure 5, where the subjective savings bias (SSH − OSH) and the health heterogeneity effect (OSH − NHH) are reported across cash-at-hand deciles for selected ages.

Looking at my replication, the plot does differ, especially when looking at the downward side. However, note that in general the signs are leading into the correct direction and are very accurate for the age-80 panel. This has a clear reason: the missing bequest system allocates additional resources over the life cycle that alter the exact savings decisions in my model. However, agents at the later stages of their lives have likely already received their bequests and thus are least affected by a missing bequest system. This explains why my replication is able to capture the results so well at age 80 but diverges at earlier ages.

Moreover, bequests are inherited on an intergenerational wealth basis; this link shifts more bequests to richer agents. Therefore, bequest inheritance is dependent on health, and thus the missing bequest system creates an additional distortion between OSH and NHH. This is why the OSH–NHH gap (hatched bars) is particularly affected, while the SSH–OSH comparison is better matched across all ages.



(a) Foltyn and Olsson (2024): Original savings rate decomposition



(b) Replication: Python implementation

Figure 5: Savings rate differences by cash-at-hand decile and health state. Upper graph shows original results and lower one replication. Hatched bars show the effect of health heterogeneity (OSH − NHH); solid bars show the additional subjective belief effect (SSH − OSH). Colors indicate health states.

4.4 General Equilibrium Summary

The capital market clears through the equilibrium interest rate r^*. Total Factor Productivity (A) is adjusted in the SSH scenario so that wage is normalized to unity. This calibrated TFP value is then used as an anchor for OSH and NHH, leading to slightly different wages.
Table 3 summarizes equilibrium values across the three belief economies and illustrates the results of the paper. The mere ordering is correct: OSH yields the lowest r^* (objective agents save more), while SSH and NHH are similar. My interest rates are roughly 10–15bp lower than the paper's, and Gini coefficients are lower (0.66 vs 0.70). The latter difference reflect the missing stochastic bequest redistribution, which adds wealth at the top of the distribution in the original model, which amplifies inequality. To compensate for the lower disposable wealth in the economy, I have to calibrate $\beta = 0.986$, versus the authors' $\beta = 0.979$, to achieve the desired target of an approximate 3.0 Capital over Output ratio as described in the paper. This mechanism also affects the interest rate, distorting the level comparison between the replicated and original interest rates.

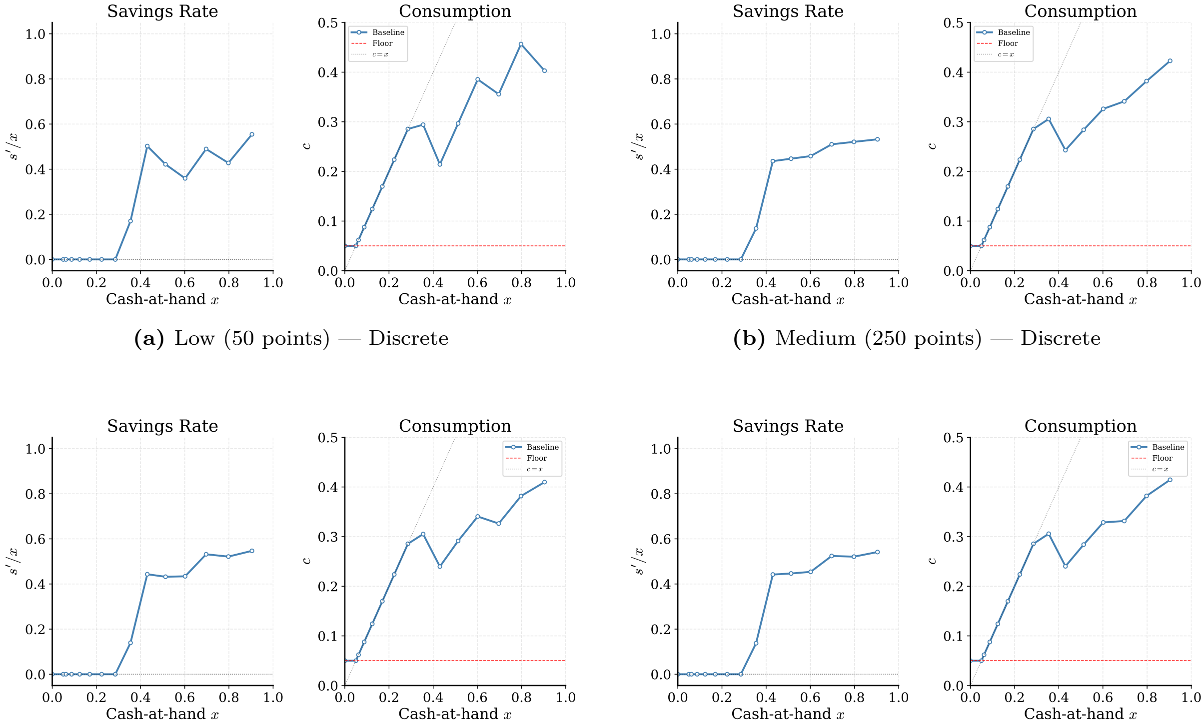
Economy	**Source**	r^*	w^*	**K/Y**	**Gini**
SSH (Subjective)	Replication	2.22%	1.0000	3.047	0.663
	Foltyn & Olsson	2.37%	1.0000	—	0.701
OSH (Objective)	Replication	2.06%	1.0068	3.088	0.661
	Foltyn & Olsson	2.19%	1.0086	—	0.702
NHH (Average)	Replication	2.20%	1.0003	3.052	0.661
	Foltyn & Olsson	2.30%	1.0032	—	0.715

Table 3: General Equilibrium Results: Replication vs. Original

For the bequest extension (Section 4.6), I re-solve SSH general equilibrium with the baseline bequest calibration, obtaining $r^* = 2.11\%$. I set $\beta = 0.96$ (versus the authors' $\beta = 0.942$) to achieve comparable K/Y ratios. This economy serves as the anchor for exploring the ϕ_2 threshold mechanism.

4.5 Hybrid GSS vs. Discrete Grid Search

To illustrate the efficiency of the hybrid solution method, I examine the savings policy at age 80 (model period 60) for an agent in the worst health, medical, and productivity states under OSH Partial Equilibrium. This agent is heavily constrained by the consumption floor $\underline{c}$, generating visible discontinuities in the policy function.
Figure 6 compares discrete grid search across three resolutions against the hybrid GSS method. The high-resolution grid (2000 points) serves as the benchmark. The low-resolution grid (50 points) produces a jumpy policy, where the optimizer creates artificial discontinuities. The medium grid (250 points) captures the general structure but still shows visible deviations from the benchmark. The hybrid GSS with 250 points achieves comparable accuracy to the 2000-point discrete grid by refining the coarse solution via continuous search.



(a) Low (50 points) — Discrete

(b) Medium (250 points) — Discrete

(c) High (2000 points) — Discrete

(d) Medium (250 points) — Hybrid GSS

Figure 6: Policy at age 80 (worst states) under OSH PE. Hybrid GSS achieves high-resolution accuracy at medium-grid cost.

Table 4 reports PE solver runtimes. The hybrid method with 250 points adds only marginal overhead versus pure discrete at the same resolution (19s vs 18s), but produces a policy visually indistinguishable from the 2000-point benchmark. Whether a 500- or 1000-point discrete grid would suffice depends on the application; the point is that hybrid GSS achieves benchmark accuracy without requiring to determine the necessary grid density ex ante.

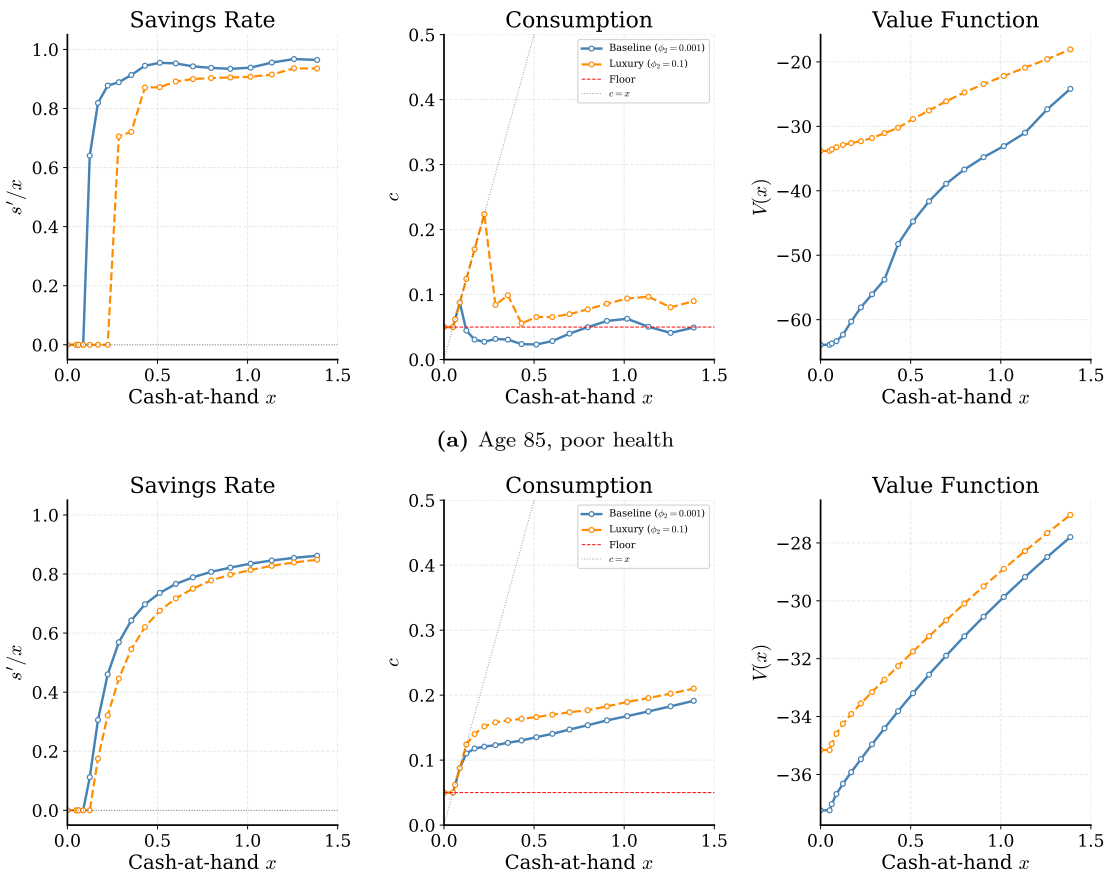
Grid	Method	Time (s)
50 points	Discrete	3.53
250 points	Discrete	17.51
2000 points	Discrete	137.16
250 points	Hybrid GSS	19.65

Table 4: Partial Equilibrium Solver Runtimes

4.6 The Bequest Motive

The paper includes a bequest motive scenario and argues that it introduces non-trivial effects on savings behavior. One effect driving savings decisions is that bequests are only realized in the death state, and thus bequest willingness is a direct function of mortality. Agents who perceive higher death probabilities place more weight on the bequest channel. The authors calibrate $\phi_1 = 11.127$, $\phi_2 = 0.001$, $\chi_b = 18.333$. The near-zero ϕ_2 makes bequests a necessary good: even small estates generate high marginal utility because of the chosen log utility. I explore what happens when ϕ_2 is raised to 0.1, making bequests more of a luxury good.

Figure 7 contrasts an age-85 agent in poor health with an age-50 agent in excellent health in the SSH economy. The older, unhealthy agent underestimates survival and overweights the bequest channel, generating a sharp savings jump. Notice that consumption drops below the consumption floor $\underline{c}$ in the baseline case. The government ensures only a minimum cash-on-hand of 0.05, and does not automatically fix consumption at this lower bound. Agents with sufficient resources choose their savings and consumption freely. The fact that the agent in the baseline scenario consumes below the floor illustrates the force of bequest marginal utility: because ϕ_2 is near zero and the agent's subjective mortality probability is high, massive weight falls on the bequest term, pulling consumption below the floor.
Raising ϕ_2 relieves this pressure. The same agent now consumes strictly above the floor, as bequest utility is modeled as more of a luxury good. The young, healthy agent shows smoother policies, in both cases survival probability dominates, and the ϕ_2 threshold matters less.

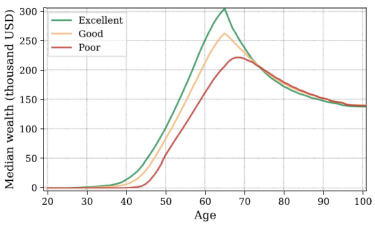


(a) Age 85, poor health

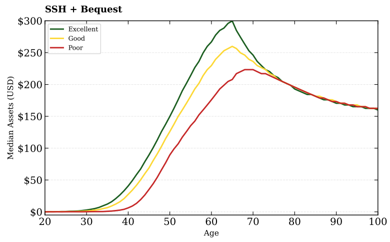
(b) Age 50, excellent health

Figure 7: Bequest policy comparison. Baseline $\phi_2 = 0.001$ (blue) vs. luxury $\phi_2 = 0.1$ (orange). The bequest motive bites hardest for old, unhealthy agents with pessimistic survival beliefs.

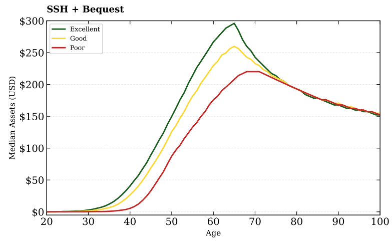
Despite these differences in policy, the ϕ_2 increase does not substantially alter median wealth dynamics. Figure 8 validates my bequest implementation against the paper and shows that both calibrations produce similar life-cycle wealth profiles.



(a) Original paper



(b) Baseline ($\phi_2 = 0.001$)



(c) Luxury ($\phi_2 = 0.1$)

Figure 8: Median wealth under SSH general equilibrium with bequests.

5 Conclusion

This paper replicates the core results of Foltyn and Olsson (2024) without modeling stochastic intergenerational bequests or a detailed government tax system. By calibrating β to match the target capital-output ratio, the model reproduces the paper's median wealth dynamics and the health-dependent savings wedge driven by subjective survival beliefs. The main limitation is wealth inequality: my Gini coefficients are lower ($\sim$0.66 vs. $\sim$0.70) because the missing stochastic bequest redistribution channel reduces wealth concentration at the top. Nonetheless, the primary mechanism of subjective survival beliefs distorting consumption-savings decisions is successfully recovered by the replication.

References

Fella, Giulio (Apr. 2014). "A generalized endogenous grid method for non-smooth and non-concave problems". In: *Review of Economic Dynamics* 17.2, pp. 329–344. ISSN: 1094-2025. DOI: 10.1016/j.red.2013.07.001. URL: http://dx.doi.org/10.1016/j.red.2013.07.001.

Foltyn, Richard and Jonna Olsson (2024). "Subjective life expectancies, time preference heterogeneity, and wealth inequality". In: *Quantitative Economics* 15.3, pp. 699–736. ISSN: 1759-7323. DOI: 10.3982/qe2016. URL: http://dx.doi.org/10.3982/QE2016.

Lam, Siu Kwan, Antoine Pitrou, and Stanley Seibert (2015). "Numba: A llvm-based python jit compiler". In: *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, pp. 1–6.

Social Security Administration (2024). *Social Security Tax Rates*. Office of the Chief Actuary. URL: https://www.ssa.gov/oact/progdata/taxRates.html (visited on 05/18/2024).